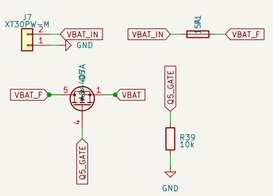
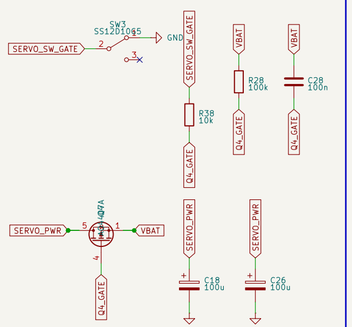
AVIONICS BOARD

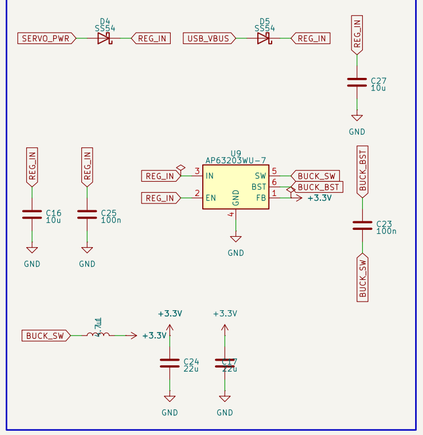
BATTERY INPUT & REVERSE PROTECTION



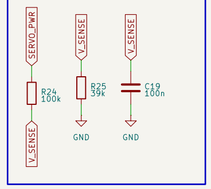
SERVO POWER SWITCH



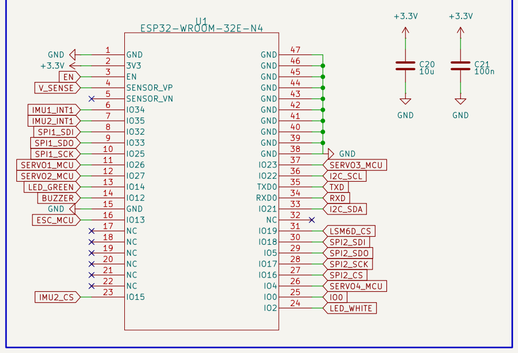
3.3V BUCK REGULATOR



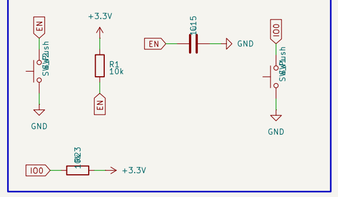
BATTERY VOLTAGE SENSE



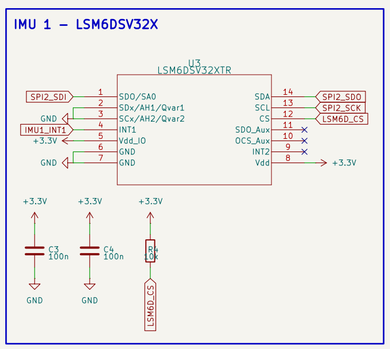
ESP32-WROOM-32E MICROCONTROLLER



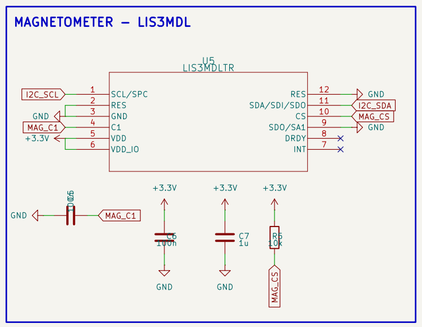
RESET & BOOT



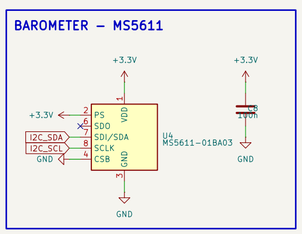
IMU 1 – LSM6DSV32X



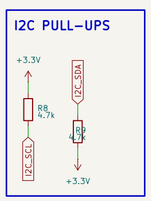
MAGNETOMETER – LIS3MDL



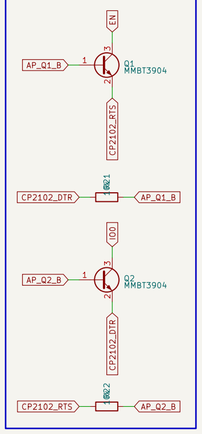
BAROMETER – MS5611



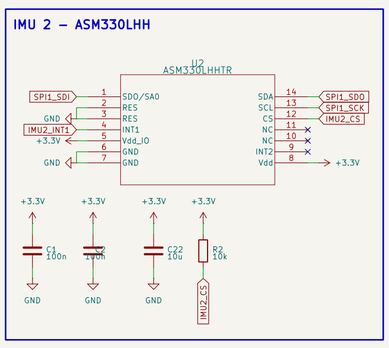
I2C PULL-UPS



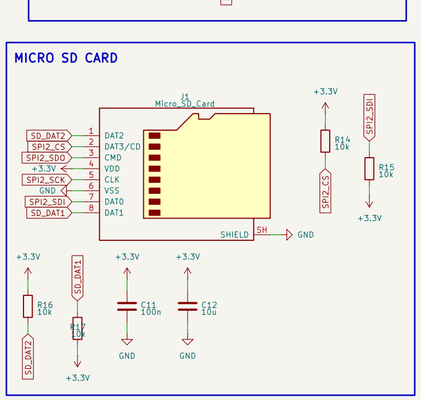
AUTO-PROGRAM



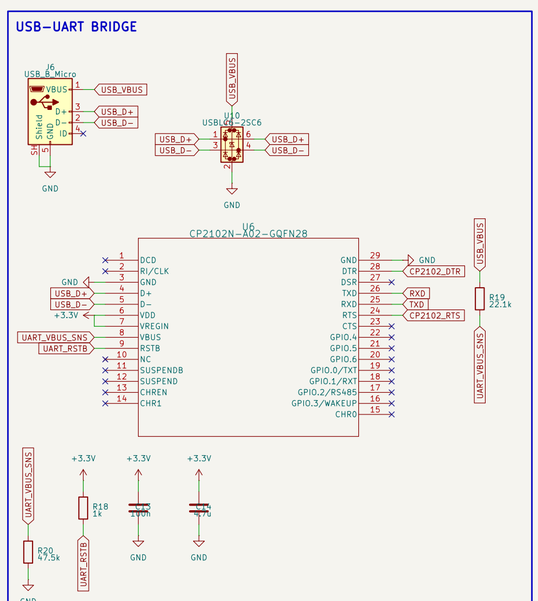
IMU 2 – ASM330LHH



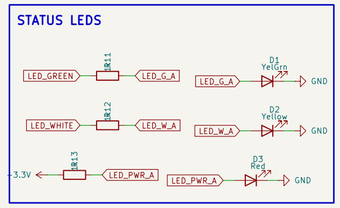
MICRO SD CARD



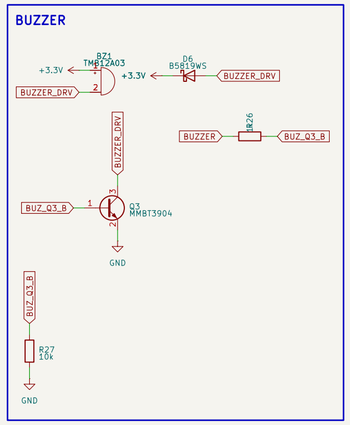
USB-UART BRIDGE



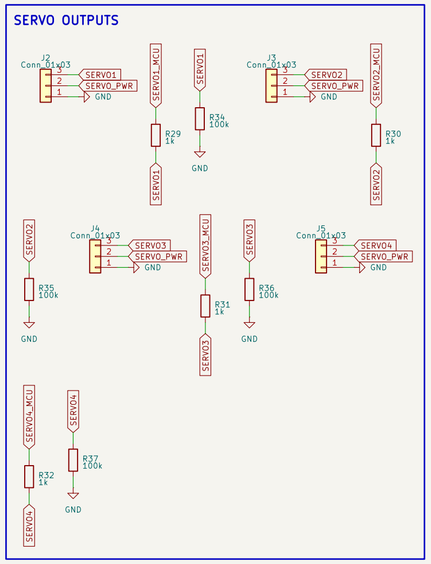
STATUS LEDS



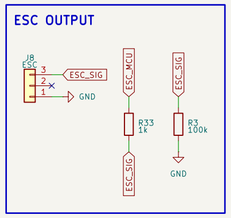
BUZZER



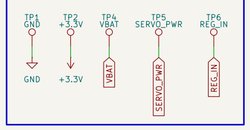
SERVO OUTPUTS



ESC OUTPUT



TEST POINTS



DESIGN NOTES

POWER
- J7 XT30PW-M (15A) -> F1 15A NANO2 fuse -> Q5 AO4407A reverse-battery PFET (gate to GND via R39) -> VBAT.
- Q4 AO4407A switches VBAT -> SERVO_PWR. SW3 pulls the gate low through R38; R28 holds Q4 off; C28 gives ~1ms soft start.
- Board rating (calculated, not yet measured): see the engineering review. Servos must be rated for 8.4V (HV).
- Electronics run from SERVO_PWR via D4 or USB VBUS via D5 -> REG_IN -> U9 AP63203 (3.8–32V in).
- V_SENSE = SERVO_PWR x 39k/(100k+39k): 2.36V at 8.4V. Inside the ESP32 11dB ADC range (150–2450mV).

INTERFACES
- Servo/ESC outputs: 1k series + 100k pull-down at each connector; outputs read LOW while the MCU is in reset.
- J8 ESC: pin1 GND, pin3 ESC_SIG (IO13). Pin2 left open so an ESC BEC cannot back-feed the 2S rail.
- No actuator on a strapping pin: IO12 = buzzer (pulled low by R26/R27). IO15 = IMU2_CS (10k pull-up). IO2 = white LED.
- IMU1 (LSM6DSV32X) INT1 -> IO34, IMU2 (ASM330LHH) INT1 -> IO35 (input-only pins, no internal pulls).
- I2C 3.3V: MS5611 0x77 (CSB=GND), LIS3MDL; 4.7k pull-ups R8/R9.
- BZ1 TMB12A03 is an ACTIVE buzzer: drive IO12 high to beep.
- USB D+/D- protected by U10 USBLC6-2SC6 (low-capacitance ESD array).

LAYOUT
- No copper pour, tracks or vias in the ESP32 antenna area.
- No tracks or vias under the MEMS sensors (ST TN0018): MS5611 kept away from the buck converter.
- High-current path on F.Cu/B.Cu power zones with solid pad connections.

2S Li-Po input: 15 A fuse, reverse protection; 4 servo + 1 ESC outputs
ESP32-WROOM-32E flight computer: dual IMU, magnetometer, barometer, microSD

James Collier

Sheet: /
File: ESP-32_Based_Avionics_Telemetry_Board.kicad_sch

Title: ESP32-Based Avionics Board

Size: A2	Date: 2026-10-28	**Rev: A**
KiCad E.D.A. 10.0.5		Id: 1/1